

§ 188. Reduced numbers.

If α is any non-zero integral or fractional number of the field Ω , then, as with the units, we consider the system of conjugate logarithms of α , by which we understand, as in § 185, the real numbers

$$\lambda_1 = \delta_1 \log |\alpha_1|, \quad \lambda_2 = \delta_2 \log |\alpha_2|, \dots, \quad \lambda_\nu = \delta_\nu \log |\alpha_\nu|, \quad (1)$$

which we also denote by $\lambda_s(\alpha)$. If we bring in the fundamental system $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}$ of units, then

we can form the linear equations, analogously to § 186, (5):

$$\begin{aligned} \xi_1 l_{1,1} + \xi_2 l_{2,1} + \dots + \xi_{\nu-1} l_{\nu-1,1} + \delta_1 \xi_\nu &= \lambda_1, \\ \xi_1 l_{1,2} + \xi_2 l_{2,2} + \dots + \xi_{\nu-1} l_{\nu-1,2} + \delta_2 \xi_\nu &= \lambda_2, \\ &\dots\dots\dots \\ \xi_1 l_{1,\nu} + \xi_2 l_{2,\nu} + \dots + \xi_{\nu-1} l_{\nu-1,\nu} + \delta_\nu \xi_\nu &= \lambda_\nu, \end{aligned} \quad (2)$$

whose determinant [§ 186, (4)] has the non-zero value

$$\begin{vmatrix} l_{1,1} & l_{2,1} & \dots & l_{\nu-1,1} & \delta_1 \\ l_{1,2} & l_{2,2} & \dots & l_{\nu-1,2} & \delta_2 \\ \dots & \dots & \dots & \dots & \dots \\ l_{1,\nu} & l_{2,\nu} & \dots & l_{\nu-1,\nu} & \delta_\nu \end{vmatrix} = nL(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}). \quad (3)$$

Accordingly the unknowns $\xi_1, \xi_2, \dots, \xi_{\nu-1}, \xi_\nu$ are determined uniquely from (2).

By adding the equations (2) one first obtains very simply

$$n\xi_\nu = \log N_a(\alpha), \quad (4)$$

and therefore ξ_ν remains the same for all numbers α that are associated with one another.

The numbers $\xi_1, \xi_2, \dots, \xi_{\nu-1}$ are called the exponents of the number α [with respect to the fundamental system $(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1})$].

From this definition it follows without further ado that the exponents of a product or of a quotient are equal to the sum or the difference of the corresponding exponents of the individual components. The exponents of a unit are rational integers, and every system of rational integers $\xi_1, \xi_2, \dots, \xi_{\nu-1}$ is also the exponent system of a unit.

The exponents of associated numbers therefore differ from one another by integers, and one can find, for each number α , an associated number α_0 whose exponents lie between 0 and 1. Such numbers are called **reduced numbers** (with respect to the fundamental system $\varepsilon_1, \dots, \varepsilon_{\nu-1}$). The reduced units are all, and only, the roots of unity present in Ω , whose number we shall denote by w . Since the two roots of unity ± 1 are contained in every field, w is at least $= 2$ and is always an even number.

If two associated numbers have the same exponent system, then they differ only by a factor which

is a root of unity. The exponent system of the reduced number α_0 derived from α is completely determined by α itself. This proves the theorem:

II. For every non-zero number α of the field Ω there exist w , and no more, reduced numbers $\rho\alpha_0$.

If another fundamental system is used in place of $(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1})$, then the exponents $\xi_1, \xi_2, \dots, \xi_{\nu-1}$ undergo an integral linear substitution of determinant ± 1 . A reduced number

can then cease to be reduced for the new fundamental system. The concept of a reduced number is therefore dependent on the choice of the fundamental system. But the number of reduced numbers derived from a given number is always the same.

By this theorem we have attained the purpose of selecting from the whole infinite system of mutually associated numbers a definite finite number of representatives.

§ 189. Bounds for the number of integral numbers of the field Ω divisible by an ideal.

Our earlier considerations have shown that every integral number of a field Ω has only a finite number of ideals as divisors. Since every integral ideal is a divisor of its norm, there are consequently also only finitely many integral numbers in Ω whose absolute norm does not exceed a given bound.

From these numbers we now again pick out a part, and ask for the number T of all **integral numbers** of the field Ω whose absolute norm does not exceed a positive magnitude t , and which are divisible by a given ideal $\mathfrak{a}$.

If we take the rational field as a preliminary example, then there T is the number of natural numbers which are smaller than t and divisible by a given rational integer m , that is, in the notation used earlier (§ 182), $T = E\left(\frac{t}{m}\right)$.

The number T is, in general, not a function of t that can be determined more exactly; it grows to infinity together with t , but since it has only integral values it changes only step by step, and hence discontinuously. What matters here is the **limit to which the ratio $T : t$ tends when t grows without bound** (which in the example cited above is $1 : m$).

This limit can be found by considerations such as those which we have already used in § 155.

We first form a basis form of the given ideal $\mathfrak{a}$:

$$\alpha = \alpha_1 x_1 + \alpha_2 x_2 + \cdots + \alpha_n x_n, \quad (1)$$

that is, a linear function of the n variables x_i having the property that, by substituting all rational integers for the x_i , from α there arise all integral numbers of the field Ω that are divisible by $\mathfrak{a}$ (§ 146). By $\alpha_{i,1}, \alpha_{i,2}, \dots, \alpha_{i,n}$ we denote the values conjugate to α_i , and form the linear substitution for the x_i :

$$\begin{aligned} y_1 &= \alpha_{1,1}x_1 + \alpha_{2,1}x_2 + \cdots + \alpha_{n,1}x_n, \\ y_2 &= \alpha_{1,2}x_1 + \alpha_{2,2}x_2 + \cdots + \alpha_{n,2}x_n, \\ &\dots\dots\dots \\ y_n &= \alpha_{1,n}x_1 + \alpha_{2,n}x_2 + \cdots + \alpha_{n,n}x_n. \end{aligned} \quad (2)$$

If A is the determinant of this system of equations,

$$A = \sum \pm \alpha_{1,1} \alpha_{2,2} \cdots \alpha_{n,n},$$

then A^2 is the discriminant of the system of functions $(\alpha_1, \alpha_2, \dots, \alpha_n)$, and therefore, by § 147, is equal to the product of the square of the absolute norm of α and the field discriminant Δ . Since the absolute norm of α is at the same time the norm of the ideal $\mathfrak{a}$ (§ 151), we put

$$A^2 = N(\mathfrak{a})^2 \Delta. \quad (3)$$

Now we can, as in § 155, interpret $x_1, x_2, \dots, x_n$ as coordinates of a point (x) in a space of n dimensions, and let there correspond to each point (x) a point (x') whose coordinates are determined by

$$x'_1 = t^{\frac{1}{n}} x_1, \quad x'_2 = t^{\frac{1}{n}} x_2, \dots, \quad x'_n = t^{\frac{1}{n}} x_n, \quad (4)$$

where t denotes an arbitrary positive magnitude. To every region S of points (x) there corresponds a region S' of points (x') . All figures in S and the corresponding figures in S' are similar, and the linear dimensions in S are to those in S' as $1 : t^{\frac{1}{n}}$.

If we think of the region S as given, then the corresponding region S' will depend on t and will increase with t . The points with integral coordinates x_i we call, as before, lattice points. The number of lattice points lying in a finite region S' is always finite, but grows with t , and shall be denoted by Z_t . Then by the theorems of § 155, (14), the volume of the region S , that is, the n -fold integral extended over all points of S ,

$$\int \int \cdots \int dx_1 dx_2 \cdots dx_n,$$

is equal to the limit of the ratio $Z_t : t$ for t increasing without bound:

$$V = \lim_{t=\infty} \frac{Z_t}{t}. \quad (5)$$

This at the same time gives the more exact precision which we must impose on the region S if our consideration is to be applicable. The region S must be such that the n -fold integral V has a definite meaning. Every lattice point is now, by (1), the image of an integral number of the field Ω divisible by $\mathfrak{a}$, and thus we can also say that Z_t is the number of integral numbers in Ω divisible by $\mathfrak{a}$ whose images lie in the region S' .

In order to bound the region S' in a suitable way, we choose a system of fundamental units of the field Ω

$$\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}$$

with the system of conjugate logarithms

$$l_{i,1}, l_{i,2}, \dots, l_{i,\nu}$$

and with regulator

$$\pm L(\varepsilon_1, \varepsilon_2, \dots, \varepsilon_{\nu-1}) = \pm \sum \pm l_{1,1} l_{2,2} \cdots l_{\nu-1,\nu-1}, \quad (6)$$

which may be denoted by L .

We now define a system of functions $\xi_1, \xi_2, \dots, \xi_\nu$ of the variables x_i by the following conditions.

Let, when the y_i have the meaning (2),

$$z_i = \delta_i \log |y_i|, \quad (7)$$

so that, when a system of rational integers is substituted for the x_i , z_i passes, by § 185, into the system of conjugate logarithms of the number α .

Then, if y_k belongs to a real field,

$$z_k = \frac{1}{2} \log y_k^2, \quad (8)$$

and if y_k, y'_k belong to an imaginary pair,

$$z_k = \log y_k y'_k, \quad (9)$$

whereby the definition is completely unambiguous.

We now determine the variables $\xi_1, \xi_2, \dots, \xi_{\nu-1}, \xi_\nu$ from the linear equations

$$\begin{aligned}\xi_1 l_{1,1} + \dots + \xi_{\nu-1} l_{\nu-1,1} + \delta_1 \xi_\nu &= z_1, \\ \xi_1 l_{1,2} + \dots + \xi_{\nu-1} l_{\nu-1,2} + \delta_2 \xi_\nu &= z_2, \\ &\dots\dots\dots \\ \xi_1 l_{1,\nu} + \dots + \xi_{\nu-1} l_{\nu-1,\nu} + \delta_\nu \xi_\nu &= z_\nu,\end{aligned}\tag{10}$$

whose determinant does not vanish.

If integral values are given to the variables x_i , then by § 188, (2), (3) the $\xi_1, \xi_2, \dots, \xi_{\nu-1}$ pass into the exponent system of the number α , and by addition of the equations (10) [§ 185, (3)] one obtains

$$\xi_\nu = \frac{1}{n} \log N_a(\alpha).\tag{11}$$

We now pass from the point (x) to the point (x') by putting $x'_i = t^{\frac{1}{n}} x_i$. Thereby y_i passes into

$$y'_i = t^{\frac{1}{n}} y_i,$$

and z_i into

$$z'_i = z_i + \frac{1}{n} \delta_i \log t.$$

If, therefore, we put z'_i in place of z_i , and denote by $\xi'_1, \xi'_2, \dots, \xi'_\nu$ the values of $\xi_1, \xi_2, \dots, \xi_\nu$ then obtained from (10), then

$$\xi'_1 = \xi_1, \quad \xi'_2 = \xi_2, \dots, \quad \xi'_{\nu-1} = \xi_{\nu-1}, \quad \xi'_\nu = \xi_\nu + \frac{1}{n} \log t.\tag{12}$$

We shall now bound the region S' by requiring that

$$0 \leq \xi'_1 < 1, \dots, 0 \leq \xi'_{\nu-1} < 1, \quad \xi'_\nu < \frac{1}{n} \log t\tag{13}$$

shall hold. In this way we achieve that the lattice points (x') lying in the region S' represent only reduced integral numbers α , and all and only those whose absolute norm is smaller than t .

Among the reduced numbers, however, there are always w associated with one another, if w denotes the number of roots of unity contained in Ω ; and if

we therefore combine these w associated numbers into one complex, then the number of these complexes is equal to the number T of all non-associated integral numbers in Ω divisible by $\mathfrak{a}$ whose absolute norm is smaller than t . Accordingly the number of lattice points lying in S' is

$$Z_t = wT.\tag{14}$$

For the bounding of the region S one obtains from (12) and (13):

$$0 \leq \xi_1 < 1, \dots, 0 \leq \xi_{\nu-1} < 1, \quad \xi_\nu < 0,\tag{15}$$

and from (5) we obtain

$$\lim_{t \rightarrow \infty} \frac{T}{t} = \frac{1}{w} \int \int \dots \int dx_1 dx_2 \dots dx_n = \frac{1}{w} V,\tag{16}$$

where the n -fold integral is to be extended over the region S determined by (15).

§ 190. Determination of the volume.

The volume V can be determined by the transformation formula for multiple integrals already adduced in § 155, according to which, if $x_1, x_2, \dots, x_n$ are functions of the variables $u_1, u_2, \dots, u_n$,

$$\begin{aligned} V &= \int \int \cdots \int dx_1 dx_2 \cdots dx_n \\ &= \int \int \cdots \int \left(\sum \pm \frac{\partial x_1}{\partial u_1} \frac{\partial x_2}{\partial u_2} \cdots \frac{\partial x_n}{\partial u_n} \right) du_1 du_2 \cdots du_n, \end{aligned} \quad (1)$$

where the functional determinant in the integral taken with respect to the variables u is to be taken with its absolute value.

This functional determinant is also equal to the reciprocal value of the determinant

$$\sum \pm \frac{\partial u_1}{\partial x_1} \frac{\partial u_2}{\partial x_2} \cdots \frac{\partial u_n}{\partial x_n}.$$

We first introduce the variables u by a linear substitution, and observe that among the fields conjugate with Ω , according to our assumption, there occur $n - \nu$ conjugate imaginary pairs

and $2\nu - n$ real ones. Accordingly, of the n functions y [§ 189, (2)]

$$y_k = \alpha_{1,k}x_1 + \alpha_{2,k}x_2 + \cdots + \alpha_{n,k}x_n \quad (2)$$

$2\nu - n$ are real and $2n - 2\nu$ are conjugate in pairs. By the n variables u we shall understand real linear combinations of the x , namely the $2\nu - n$ real y and the $2n - 2\nu$ components of the conjugate imaginary pairs of the y , so that, if y, y' is an imaginary pair,

$$2u_1 = y + y', \quad 2iu_2 = y - y',$$

or

$$y = u_1 + iu_2, \quad y' = u_1 - iu_2 \quad (3)$$

is put. If in the functional determinant of the y one applies twice, to each pair of conjugate imaginary rows, the elementary theorem on determinants concerning the addition of rows (Vol. I, § 22), then there results

$$\sum \pm \frac{\partial u_1}{\partial x_1} \frac{\partial u_2}{\partial x_2} \cdots \frac{\partial u_n}{\partial x_n} = \frac{1}{(-2i)^{n-\nu}} \sum \pm \frac{\partial y_1}{\partial x_1} \frac{\partial y_2}{\partial x_2} \cdots \frac{\partial y_n}{\partial x_n} = \frac{A}{(-2i)^{n-\nu}},$$

where A has the meaning fixed in § 189, (3).

The field discriminant is positive or negative according as the number of conjugate imaginary pairs, that is, $n - \nu$, is even or odd; consequently, if $\pm\Delta$ denotes the absolute value of the field discriminant, one obtains from § 189, (4), (16):

$$\lim \frac{T}{t} = \frac{2^{n-\nu}}{wN(\mathfrak{a})\sqrt{\pm\Delta}} \int \int \cdots \int du_1 du_2 \cdots du_n. \quad (4)$$

It therefore remains only to treat the integral

$$U = \int \int \cdots \int du_1 du_2 \cdots du_n.$$

From the equations § 189, (10), for every system of values of the variables $\xi_1, \xi_2, \dots, \xi_\nu$ satisfying the conditions (15), one obtains a definite finite system of values for $z_1, z_2, \dots, z_\nu$,

and from these, by § 189, (7), for every y a non-zero absolute value. In the case of an imaginary pair (3), put

$$u_1 = e^{\frac{1}{2}z} \cos \vartheta, \quad u_2 = e^{\frac{1}{2}z} \sin \vartheta, \quad (5)$$

$$y_1 y_1' = u_1^2 + u_2^2 = e^z. \quad (6)$$

Then, when ϑ is taken within the bounds

$$0 \leq \vartheta < 2\pi,$$

there results, for fixed z , a corresponding pair of values u_1, u_2 for each value of ϑ . Such an angle ϑ belongs to every imaginary pair.

Thus to every point of the region S there belongs one and only one system of values of the variables $\xi_1, \xi_2, \dots, \xi_\nu, \vartheta \dots$ within the bounds

$$0 \leq \xi_1 < 1, \dots, 0 \leq \xi_{\nu-1} < 1, \quad \xi_\nu < 0, \quad 0 \leq \vartheta < 2\pi, \dots \quad (7)$$

Conversely, if any system of values of the variables ξ, ϑ is given within the bounds (7), then from it one obtains uniquely the corresponding components u_1, u_2 of an imaginary y , but of the real y only the absolute values; hence to each real y either of the two signs can still be given. The number of these possible determinations is $2^{2\nu-n}$. If one of these determinations is selected, the variables x are uniquely determined and are finite for every finite system of values of the ξ, ϑ . For ξ_ν decreasing toward $-\infty$, the values of the y will in part converge to zero.

Accordingly the region S decomposes into $2^{2\nu-n}$ subregions $S_1, S_2, \dots$, each of which is characterized by a definite combination of signs of the real $y_1, y_2, \dots$; and accordingly the integral U decomposes into just as many partial integrals $U_1, U_2, \dots$,

$$U = U_1 + U_2 + \dots$$

In carrying out the integration in one of these components, which all turn out to be of the same value, we begin with an imaginary pair, if one is present.

If we regard u_1, u_2 as rectangular coordinates in a plane, then by (5) we can regard $e^{\frac{1}{2}z}, \vartheta$ as polar coordinates, and obtain

$$\int \int du_1 du_2 = \frac{1}{2} \int \int e^z dz d\vartheta,$$

where the integration with respect to ϑ , which extends from 0 to 2π , can be carried out. Thus one finds

$$\int \int du_1 du_2 = \pi \int e^z dz.$$

One now changes the order of integration and puts first any second imaginary pair that may still be present, which is again treated in the same way.

The different components $U_1, U_2, \dots$ differ by the signs of the u corresponding to the real y . We take the sign so that $\pm u$ becomes positive, and put

$$z = \log(\pm u), \quad \pm du = e^z dz. \quad (8)$$

Thus for the different $U_1, U_2, \dots$ we obtain the same expression by a ν -fold integral,

$$U_1 = \pi^{n-\nu} \int \int \dots \int e^{z_1+z_2+\dots+z_\nu} dz_1 dz_2 \dots dz_\nu, \quad (9)$$

if we take $dz_1, dz_2, \dots, dz_\nu$ to be positive.

But by (6) and (8) the variables z_k are here no others than those in § 189, (8), (9), (10), and hence

$$z_1 + z_2 + \dots + z_\nu = n\xi_\nu.$$

If we now apply the transformation formula (1) to the ν -fold integral (9), in order to introduce the variables $\xi_1, \xi_2, \dots, \xi_\nu$, then we have to form the determinant

$$\sum \pm \frac{\partial z_1}{\partial \xi_1} \frac{\partial z_2}{\partial \xi_2} \dots \frac{\partial z_\nu}{\partial \xi_\nu},$$

which by § 188, (3), agrees in absolute value with nL , that is, with the n -fold value of the field regulator L ; and therefore

$$U_1 = \pi^{n-\nu} nL \int_0^1 d\xi_1 \dots \int_0^1 d\xi_{\nu-1} \int_{-\infty}^0 e^{n\xi_\nu} d\xi_\nu = \pi^{n-\nu} L$$

and

$$U = 2^{2\nu-n} \pi^{n-\nu} L.$$

Finally, therefore, (4) gives the theorem:

- 1. If T denotes the number of non-associated integral numbers divisible by $\mathfrak{a}$ whose absolute norm is smaller than t , then**

$$\lim_{t=\infty} \frac{T}{t} = \frac{2^\nu \pi^{n-\nu} L}{w N(\mathfrak{a}) \sqrt{\pm \Delta}} = \frac{g}{N(\mathfrak{a})}, \quad (10)$$

where g is a positive number completely determined by the nature of the field Ω , namely:

$$g = \frac{2^\nu \pi^{n-\nu} L}{w \sqrt{\pm \Delta}}. \quad (11)$$